

13. Generate a parse tree for a given sentence using a context-free grammar using python program.

Python Code

```
sentence = input("Enter sentence: ").split()

if sentence == ["John", "runs"]:

    print("Parse Tree")

    print("    S")

    print("   /\\"")

    print("  NP VP")

    print(" |  |")

    print("John runs")

else:

    print("Sentence not found in grammar")
```

Input

John runs

Output

Parse Tree

```
    S
   /\
  NP VP
 |  |
John runs
```

John runs

